

Question Paper

Exam Date & Time: 21-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

DATA PREPROCESSING AND VISUALIZATION [DS_3154]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A movie streaming platform uses an AI-based recommendation system to decide whether to recommend a new movie (action α_1) or not recommend it (action α_2) to a user. The system estimates that the probability the user will enjoy the movie (ω_1) is 0.4, and the probability the user will not enjoy it (ω_2) is 0.6. If the system recommends a movie the user dislikes, it results in a loss of 30 units (due to poor engagement), while failing to recommend a movie the user would have liked causes a smaller loss of 10 units (missed opportunity). Correct recommendations cause no loss. Using Bayesian Decision Theory, compute the conditional risk for both actions (recommend or not recommend), identify the minimum-risk decision, and explain what this result implies for optimizing the platform's recommendation strategy. (5)
- A)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) A marketing analyst wants to check whether type of advertisement influences product preference among consumers. The following data were collected from 100 people: (3)

Product Preference	Online Ad	TV Ad	Total
Liked Product	30	50	80
Did Not Like	10	10	20
Total	40	60	100

Using the Chi-Square test of independence,

1. Calculate the expected frequency for each cell.
2. Write the formula used to compute the expected frequencies.
3. Based on the data, state whether advertisement type and product preference appear to be independent (no need to calculate χ^2 value).

(Use 5% level of significance; χ^2 table value at $df = 1$ is 3.84)

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) List any two components of the visualization design framework. (2)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 2) A hospital is developing a predictive model to identify patients at high risk of diabetes using data collected from multiple clinics. The combined dataset includes missing entries in blood glucose levels, duplicate patient IDs across different clinics, inconsistent data formats (some glucose values in mg/dL and others in mmol/L), and continuous attributes such as age and BMI. Additionally, categorical data such as gender and region come from different coding systems across clinics. As a data scientist, analyze the preprocessing challenges in this scenario and explain how you would apply data cleaning, data integration, data transformation, and data discretization techniques to prepare the data for analysis. Provide clear justification for each step you choose and describe how these processes improve data quality and model performance. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Evaluate"
- B) What are the key elements of a computer-based visualization system? (3)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- C) What is a discriminant function in pattern classification? (2)
- Course Outcome: "CO1" / Cognitive Level: "Remember"
- 3) With suitable examples, explain Task Abstraction in visualization, focusing on *actions* (analyze, search, query) and *targets* (attributes, trends, outliers). (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) What is a minimum error rate classifier? Derive the corresponding decision rule. (3)
- Course Outcome: "CO1" / Cognitive Level: "Understand"
- C) What are visual encoding principles? Give one example each for color and size encoding. (2)
- Course Outcome: "CO3" / Cognitive Level: "Remember"
- 4) Explain the concepts of marks and channels in data visualization. Discuss the different types of marks and channels, and their roles in effectively encoding data. Give one example of each. (5)
- A) Course Outcome: "CO4" / Cognitive Level: "Application"
- B) What are visual channels? Give examples of how they are used in visualization. (3)
- Course Outcome: "CO3" / Cognitive Level: "Understand"
- C) The following data are given: $X = [1, 2, 3]$, $Y = [2, 4, 6]$. Determine the relationship between X and Y using simple linear regression (2)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 5) A fitness center wants to classify a new member as either "Fit" or "Unfit" based on their Age and Weekly Exercise Hours. (5)
- A) The data for six existing members is shown below.

Member	Age	Exercise Hours	Class
A	25	6	Fit
B	30	5	Fit
C	45	2	Unfit
D	50	1	Unfit
E	28	4	Fit
F	35	3	Unfit

Using $K = 3$ and Euclidean distance, classify a new member P(40 years, 2 hours).

1. Compute the distance of P from all members.
2. Identify the 3 nearest neighbors.
3. Predict the class label of P using majority voting.
4. Write one advantage and one limitation of the KNN algorithm.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Compare and analyze the use of explicit spatial layouts (maps, geographic plots) versus implicit algorithmic layouts (force-directed graphs, treemaps) for network data visualization (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) A bank is analyzing customer data to predict Loan Approval based on Credit Score: (2)

Credit Score	Loan Status
Good	Approved
Good	Approved
Poor	Approved
Poor	Rejected

Calculate the Gini Index for the attribute *Credit Score*.

Course Outcome: "CO2" / Cognitive Level: "Application"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	17
3	3	CO 3	15
4	4	CO 4	8
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	2	4(8%)
2	Understand	3	8(16%)
3	Application	4	12(24%)
4	Analysis	5	21(42%)
5	Create	1	5(10%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)